

Highlights

Factored Interaction Fields: Learning Joint-Anchored Hand-Object Proximity Through Its Geometric Arguments

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- Interaction fields predicted through their arguments: joints, object pose, mesh
- Factored model beats a matched direct regressor on two held-out subject pairs
- Gains concentrate in far-field, occluded and out-of-view joints, reaching 21 mm
- The gain is carried by geometric supervision, not by inference-time fusion
- The advantage does not transfer to HOT3D and vanishes under joint training